

Stage	Function	Description	Verification Method	Steps	Result	Notes
Before Assembly	QUALITY CONTROL	PCB Quality Control	Inspection	Check the PCB is has been manufactured to our expected quality	PASS	
POWER						
Voltage Regulation and Protection	POWER DISTRIBUTION	Power and Ground are Isolated	Test	Use a multimeter to read that the resistance between power and ground is an open loop	PASS	
Before Assembly	ASSEMBLY	Solder commissioning input and 12V_RAW status LED and test point	Inspection	Visually inspect solder joints, measure continuity between net nodes then supply 12V at the input and observe 12V_RAW LED on	PASS	
During Assembly - Unpowered	ASSEMBLY	Solder reverse polarity proection, fuse and status LED + TP	Inspection	Visually inspect solder joints, measure continuity between net nodes then supply 12V at the input and observe 12V_PROT LED on	PASS	
During Assembly - Unpowered	ASSEMBLY	Solder 5V regulator circuit and status point	Inspection	Visually inspect solder joints, measure continuity between net nodes then supply 12V at the input and observe 5V LED on	PASS	
During Assembly - Unpowered	POWER DISTRIBUTION	Check all supply voltages	Test	Supply the commissioning port with 12V and measure all of the supply voltages to be true.	PASS	
TEENSY						
During Assembly - Unpowered	ASSEMBLY	Solder teensy pins	Inspection	Visually inspect solder joints, measure continuity between net nodes	PASS	
During Assembly - Powered	Teensy power	Check power	Inspection	Measure teensy supply voltage	PASS	
SHUTDOWN						
During Assembly - Unpowered	ASSEMBLY	Solder shutdown fuse and diode	Inspection	Visually inspect solder joints, measure continuity between net nodes	PASS	
During Assembly - Unpowered	ASSEMBLY	Solder Input Protection & Putdown for Teensy Input	Inspection	Visually inspect solder joints, measure continuity between net nodes	PASS	
During Assembly - Powered	SHUTDOWN	Validate SHUTDOWN	Test	Supply the commissioning port with 12V, measure ST_Start to be 12V. Observe SD_BOTS LED on only when a closed circuit is between SD start and SD BOTS (simulating the switch). Measure SD_Bots as 12V and both teensy signals as below 3.3 but still a teensy high (zener diode soft knee may limit this to ~2.2 V)	PASS	
SENSORS						
During Assembly - Powered	ASSEMBLY	Solder PTC fues	Inspection	Supply 12V, Measure 5V at the 4 TPS and BPS supply pins	PASS	
During Assembly - Unpowered	ASSEMBLY	Solder the 4 input filters checking for unpopulated components.	Inspection	Supply each sensor input with 5v and measure the teensy inputs below 3.3V	PASS	
During Assembly - Powered	SENSORS	Replicate PEN V1 Sensor Testing results with singal generator and oscilloscope	Analysis	Using the report as a guide, replicate the results with the voltage ranges for each sensor	PASS	
PRESSURE						
During Assembly - Unpowered	ASSEMBLY	Solder voltage dividers and input protection for OCSC and Tresh	Test	Supply Pressure OCSC and Thresh with 5v checking 3.3v for teensy inputs	PASS	
During Assembly - Unpowered	ASSEMBLY	Solder OCSC circuit	Inspection	Solder reference voltages, comparitors etc. Do not forget protection caps for Ics. Visually inspect solder joints, measure continuity between net nodes	PASS	
During Assembly - Powered	OCSC	Test OCSC fault for each sensor	Test	Observe indicator LED on and PRESSURE_OCSC signal at 5V when healthy and off + 0V for faulted	PASS	
During Assembly - Unpowered	ASSEMBLY	Solder threshold check circuit with protection caps.	Inspection	Visually inspect solder joints, measure continuity between net nodes	PASS	
During Assembly - Powered	Threshold	Set pressure front and back reference voltages with the potentiometers. (PEN V1 values should be checked to avoid unnecessary tuning while on the car)	Inspection	Supply critical brake sensor voltages above and below reference. Measure PRESSURE_THRESH and LED status to be Low under "heavy braking"	PASS	
CAN						
During Assembly - Unpowered	ASSEMBLY	Solder can transeiver circuit	Inspection	Visually inspect solder joints, measure continuity between net nodes	PASS	
Post Assembly	CAN	Read CAN data	Demonstration	Upload PEN code to teensy, use can king + kvaizer to read signals from teensy. i.e. heartbeat, pressures and apps	PASS	
BENCH TESTING						
During Assembly - Unpowered	ASSEMBLY	Solder DT connectors	Inspection	Visually inspect solder joints, measure continuity between net nodes	PASS	
Post Assembly	Code integration	Scaling checks	Demonstration	Using the teensy serial monitor and a variable power supply on the dt sensor pins, check the scaling of the throttle and brake pressure ouputs. (check sensor voltage ranges from the testing report)	PASS	
Final Bench Test	Code integration	BOTS	Demonstration	Demonstrate bots being read on the teensy	PASS	
Final Bench Test	Code integration	Thresh	Demonstration	Repeat the PRESSURE_THRESH & PRESSURE_OCSC test while using the DT connectors and reading the teensy values.	PASS	
Final Bench Test	CAN messages	Check outputted values	Demonstration	Repeat the prior 2 tests while reading the can bus traffic	PASS	
INTEGRATION						
Integration Test	POWER DISTRIBUTION	Plug DEN DT into PEN and turn on car LV	Demonstration	Observe the board receive power from the car	PASS	
Integration Test	SD_BOTS	Connect the BOTS loom to PEN	Demonstration	Demonstrate the switch deactivating the LED and monitor the serial channel on the TEENSY or the can read out for BOTS	PASS	
Integration Test	SENSORS	Connect sensor loom	Demonstration	With HV disabled. Test the pedals reading correct values over CAN with correct range	PASS	
Integration Test	BSPD, APPS TRAIL	Test and tune pressure Thresh, observe Pressure OCSC off when sensors not plugged in	Demonstration	Apply pressure to the brake and check threshold. Ensure apps trail is activate before BSPD when on both pedals.	PASS	
Integration Test	APPS GATED	Spin the wheels	Demonstration	Turn on HV, activate RTD, test spinning the wheels and then braking.	PASS	